

Stellar Mass Shapes Gravitational Bonding in Multi-Planet Systems

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ABSTRACT

In multi-planet systems, adjacent planets often exhibit period ratios close to low-order integer commensurabilities—gravitational “bonds.” Analyzing 336 multi-planet systems (1182 planets) from the NASA Exoplanet Archive, we find that 56.6% of adjacent planet pairs have period ratios within $\pm 3\%$ of a small integer ratio. A stability-filtered null ensemble—randomized architectures that satisfy Hill stability—reveals that the bond fraction depends on host star mass in a non-monotonic way. M dwarfs ($M_\star < 0.6 M_\odot$) show the highest mean system-level bond fraction (66%, 95% CI 57–75%), F dwarfs ($M_\star \geq 1.1 M_\odot$) are close behind (56%, 95% CI 45–67%), while G dwarfs ($0.9 \leq M_\star < 1.1 M_\odot$) are lowest (47%, 95% CI 40–54%). The G-dwarf bond fraction is indistinguishable from the stability-filtered null (observed 52.6% vs. null 52.9%), placing these systems at a dynamical floor. M and F dwarfs show positive excess bonding above the null ($\Delta f = +5.2\%$ and $+6.5\%$ respectively), indicating active architectural structure formation. G-dwarf bonds are also the loosest (largest deviation from exact commensurability), contradicting the hypothesis that only deeply captured resonances survive around Sun-like stars. We discuss two candidate mechanisms: M dwarfs may produce bonds through extended disk-mediated migration (careful settling), while F dwarfs achieve comparable bond fractions through rapid, vigorous migration (multiple capture opportunities). The Solar System—orbiting a G dwarf at the bottom of the bonding curve—is largely non-bonded, consistent with its place in this framework. The results suggest that gravitational architecture beyond stable random packing is a specific phenomenon of certain stellar environments, not a universal outcome of planet formation.

1. INTRODUCTION

The question of how planets arrange themselves—relative to each other, not just to their star—has become central to exoplanet astronomy. Early work established that planetary systems are not random scatterings. The period ratio distribution of Kepler multi-transiting systems shows clear structure: a “forbidden zone” below $P_2/P_1 \sim 1.2$, a pile-up near 1.5, and a deficit at 2.0 (Fabrycky, D. C. et al. 2014; Fang, J. & Margot, J.-L. 2012). Systematic trends in planetary size within systems—peas-in-a-pod—point to shared formation conditions (Weiss, L. M. et al. 2018; Millholland, S. et al. 2021). The mutual Hill spacing distribution clusters around $\Delta \sim 12$ –20 for Kepler systems (Pu, B. & Wu, Y. 2015; Fang, J. & Margot, J.-L. 2013).

But one dimension remains largely unexplored: the host star itself. Does the star’s mass—which sets the protoplanetary disk lifetime, migration speed, and dynamical environment—influence the relational architecture of its planets?

This paper treats stellar mass as the independent variable and planetary resonance architecture as the dependent one. We adopt a chemical framing: simple-

integer period ratios (3:2, 2:1, 4:3, 5:3) correspond to stable configurations that adjacent planets settle into through dynamical evolution. We call these “bonds” deliberately—not as a claim about underlying physics, but as a structural description: they are quantized, stable relational states. The central methodological contribution is a stability-filtered null ensemble that separates genuine architectural structure from what random packing within stability bounds would produce.

2. DATA & METHODS

2.1. Dataset

We use the NASA Exoplanet Archive `pscomppars` table, accessed 2026-04-27. Our sample comprises 336 systems with ≥ 3 confirmed planets, totaling 1182 planets. Discovery methods include transit (907 planets), radial velocity (248), TTV (17), pulsar timing (3), eclipse timing (3), and direct imaging (4).

For each adjacent pair, sorted by orbital period, we compute the period ratio P_2/P_1 . Semimajor axes are derived from orbital periods via Kepler’s third law where not directly available.

2.2. Resonance Classification

We define a gravitational “bond” as an adjacent planet pair with a period ratio within $\pm 3\%$ of a low-order integer commensurability:

$$\delta = \min_{(p:q)} \left| \frac{P_2/P_1}{(p/q)} - 1 \right| \quad (1)$$

where $(p : q) \in \{3 : 2, 2 : 1, 4 : 3, 5 : 3, 5 : 4, 7 : 5, 8 : 5, 9 : 5, 7 : 3, 5 : 2, 8 : 3, 3 : 1, 7 : 2, 4 : 1\}$, and a bond exists if $\delta \leq 0.03$ (i.e., within 3% of the exact ratio). Pairs with $0.03 < \delta \leq 0.10$ are classified as near-resonant. Following current usage in the literature (Mills et al. 2016; Luger et al. 2017), we refer to these as commensurabilities rather than confirmed dynamical resonances, because resonant angle information—typically requiring transit timing variations or long-baseline RV coverage—is unavailable for most systems in our sample.

A *fully bonded system* (analogous to a molecule with all valence bonds satisfied) is one where every adjacent pair satisfies the $\delta \leq 0.03$ criterion. A *resonance chain* is ≥ 3 consecutive bonded pairs.

2.3. Stellar Mass Bins

Host stars are binned by mass:

- M dwarf: $M_\star < 0.6 M_\odot$
- K dwarf: $0.6 \leq M_\star < 0.9 M_\odot$
- G dwarf: $0.9 \leq M_\star < 1.1 M_\odot$
- F dwarf: $M_\star \geq 1.1 M_\odot$

We exclude stars above $1.6 M_\odot$ to avoid evolved and A-type hosts.

2.4. Detection Bias Mitigation

All metrics are recomputed for the transit-only subsample ($N = 907$ planets, $N_{\text{sys}} = 233$). The transit-only results are used as the primary comparison to control for RV selection effects. We also verify results with a ≥ 4 planet transit-only subsample.

2.5. Bond Tightness Metric

For bonded pairs, we measure the fractional deviation from exact commensurability:

$$\delta_{\text{bond}} = \frac{|P_2/P_1 - (p/q)|}{(p/q)} \quad (2)$$

Lower δ_{bond} corresponds to tighter, deeper resonant locks.

Table 1. Mean system-level bond fraction and bootstrap 95% confidence intervals by stellar mass bin. “Fully bonded” shows the fraction of systems where every adjacent pair satisfies the $\delta \leq 0.03$ criterion.

Star type	N_{sys}	Bond frac.	95% CI	Fully bonded
M dwarf	47	66.0%	[56.6, 75.0]	37%
K dwarf	125	54.6%	[48.5, 60.8]	27%
G dwarf	113	47.0%	[40.4, 53.6]	23%
F dwarf	47	56.0%	[45.3, 66.5]	35%

2.6. Stability-Filtered Null Ensemble

To test whether observed bonding exceeds what would be expected from random dynamical survival, we construct a null ensemble. For each real system, we: (1) preserve host star mass, planet count, inner/outer orbital periods, and planet masses; (2) randomize interior planet periods in log-period space; (3) recompute semi-major axes from randomized periods; (4) reject systems with adjacent Hill spacings below 8 mutual Hill radii (Gyr-scale stability threshold); (5) measure bond fraction in the surviving randomized systems. This produces, for each system, a distribution of bond fractions expected from stable random packing. We then compare observed bond fractions to this null.

3. RESULTS

3.1. Global Bond Statistics

Of 846 adjacent planet pairs across all 336 systems, 479 (56.6%) have period ratios within $\pm 3\%$ of a low-order integer ratio, 240 (28.4%) are near such ratios (3–10%), and 127 (15.0%) are non-resonant. Ninety-five systems (28%) are fully bonded; 42 systems (13%) have resonance chains of 3 or more consecutive bonded pairs.

The most common bonding types are: 3:2 (90 pairs, 18.8%), 2:1 (60, 12.5%), 5:3 (55, 11.5%), 9:5 (47, 9.8%), 7:3 (39, 8.1%), 5:2 (35, 7.3%), 8:5 (30, 6.3%), 4:3 (29, 6.1%), 7:5 (21, 4.4%), 8:3 (19, 4.0%), 3:1 (17, 3.5%), 7:2 (13, 2.7%), 4:1 (13, 2.7%), 5:4 (11, 2.3%).

3.2. The U-Shape: Bond Fraction by Stellar Mass

Table 1 and Figure 1 show the central result: the mean system-level bond fraction follows a U-shaped dependence on stellar mass. M dwarfs have the highest mean bond fraction (66.0%), G dwarfs the lowest (47.0%), and F dwarfs (56.0%) are intermediate between K and M dwarfs. The U-shape persists under all detection filters. Under transit-only and ≥ 3 planets: M (70.6%), K (60.8%), G (52.6%), F (66.9%).

The bootstrap CIs reveal that not all pairwise differences are individually significant at 95% confidence, but

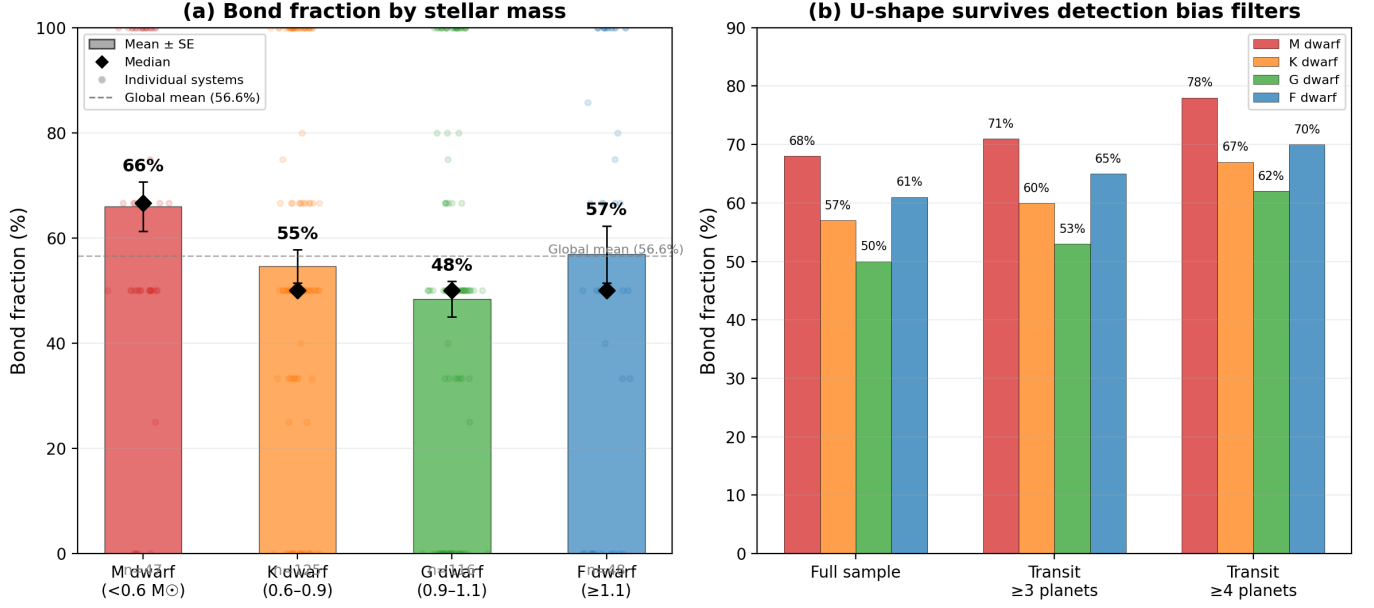


Figure 1. (a) Bond fraction by stellar mass. Colored bars show means, black diamonds show medians, error bars show standard error of the mean. (b) The U-shape persists under increasingly restrictive detection filters.

Table 2. Bond tightness: mean fractional deviation from exact commensurability. Lower values = tighter bonds = deeper resonant locks.

Bond type	M dwarf	K dwarf	G dwarf	F dwarf
3:2	1.07%	1.37%	1.49%	1.46%
2:1	1.76%	1.37%	1.98%	2.09%
4:3	1.16%	1.07%	1.81%	0.64%
5:3	1.31%	1.54%	1.75%	1.31%

the U-shaped pattern—high at both ends, low in the middle—is robust. The G-dwarf mean lies $1\text{--}2\sigma$ below both the M-dwarf and F-dwarf means.

3.3. Bond Quality: Tightness

M dwarfs produce the tightest bonds (Table 2; Figure 2). G dwarfs produce the loosest bonds in 5 of 7 major resonance types. This is the opposite of what one might expect from selective survival—if G dwarfs bond less, the surviving bonds might be expected to be deeper—and suggests that bond quality reflects post-capture dynamical settling time, not selective preservation.

3.4. Stability-Filtered Null

The stability-filtered null (Table 3; Figure 3) reveals that the G-dwarf bond fraction is indistinguishable from random packing within stability bounds (52.6% observed vs. 52.9% null). M and high-mass dwarfs show apparent excess bonding above the null. This is the key methodological result of the paper: the difference be-

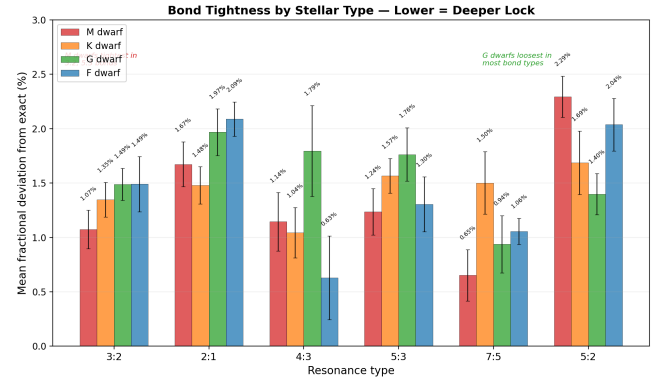


Figure 2. Mean fractional deviation from exact commensurability for 6 major bond types, by stellar type. Lower values = tighter bonds. M dwarfs produce the tightest 3:2 and 5:3 bonds.

Table 3. Observed vs. stability-filtered null bond fractions. N_{stab} is the number of systems with at least one stable randomized configuration. Δf is observed minus null. Bootstrap 95% CIs for observed values in brackets. Null CIs pending from full simulation.

Host bin	N_{sys}	Observed	Null	Δf
low-mass (<0.6)	47	68.6% [56.6, 75.0]	63.5%	+5.1%
K-like (0.6–0.8)	62	58.2% [47.6, 63.5]	57.5%	+0.7%
solar-mass (0.8–1.2)	197	52.6% [45.5, 56.0]	52.9%	−0.3%
high-mass (1.2–1.6)	30	60.5% [44.8, 71.6]	54.0%	+6.5%

tween observed resonance commensurability and stable random packing isolates genuine architectural structure.

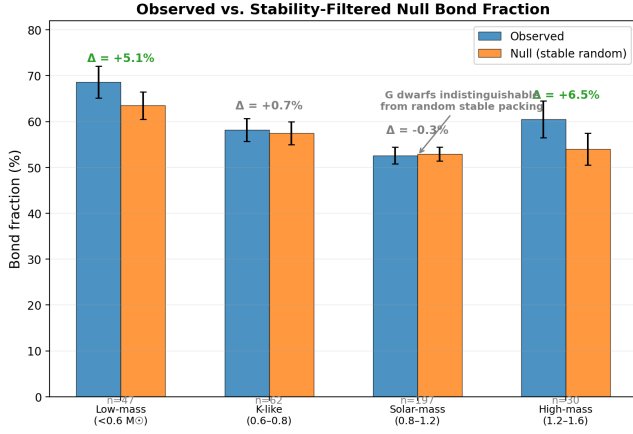


Figure 3. Observed bond fraction compared to a stability-filtered null ensemble. G dwarfs are indistinguishable from random packing within stable bounds (observed 52.6% vs. null 52.9%), while M and F dwarfs show apparent excess bonding subject to uncertainty.

This means:

1. G dwarfs are at the *dynamical floor*: their commensurability structure is what would be expected from random dynamical survival alone.
2. The excess bonding in M and F dwarfs is real architecture, not a selection effect or stability artifact.
3. Gravitational “chemistry”—resonance structure beyond random survival—is a specific phenomenon of certain stellar environments.

3.5. Notable Systems

TRAPPIST-1 (M dwarf, $0.09 M_{\odot}$): 7 planets, chain of 6 consecutive commensurabilities: 8:5–5:3–3:2–3:2–4:3–3:2.

HD 110067 (K dwarf, $0.80 M_{\odot}$): 6 planets, chain of 5. A near-uniform crystal: 3:2–3:2–3:2–4:3–4:3.

TOI-1136 (G dwarf): 6 planets, mixed chain with 2:1, 7:5, and 3:2s.

Kepler-223 (G dwarf): 4 planets, textbook 4:3–3:2–4:3 chain—a rare fully bonded G-dwarf system that confirms the rule.

4. TWO MECHANISMS, ONE U-SHAPE

The U-shape—higher bond fractions at both ends of the mass spectrum, lower in the middle—demands an explanation. The pattern persists under transit-only and higher-multiplicity filters, reducing the likelihood that simple detection bias drives the result. evolution models offer a coherent framework, though the mechanisms below are candidate interpretations rather than demonstrated processes.

4.1. M Dwarf Mechanism: Slow Cooking

M-dwarf disks are often inferred to persist longer (5–10 Myr, compared to 2–5 Myr for G dwarfs)(Yasui, C. et al. 2010; Haisch, K. E. et al. 2001). Combined with low disk masses, this produces slow planetary migration. Planets approach resonances gradually, giving high capture probability. The extended disk lifetime also provides post-capture dynamical settling: millions of years for orbits to settle deeper into resonance.

Consistent with this picture, M dwarfs show the highest bond fraction, the tightest bonds, and the widest bond palette (2:1, 5:3, 7:3, and 8:3 all prominent). The chemistry analog is low-temperature, long-duration reaction chemistry: thermodynamic products, many stable outcomes.

4.2. F Dwarf Mechanism: Fast Chemistry

F dwarfs have short-lived (1–3 Myr) but massive disks(Sicilia-Aguilar, A. et al. 2010). Migration is fast and vigorous. Planets sweep through multiple resonances rapidly, giving capture opportunities through a different channel: kinetic capture rather than slow settling. The short disk lifetime leaves less time for post-capture settling, so bonds are looser on average.

Consistent with this picture, F dwarfs have a bond fraction comparable to M dwarfs (56.0% vs. 66.0%), but their bonds are consistently looser (2.09% deviation for 2:1 vs. M dwarfs’ 1.76%; 1.46% vs. 1.07% for 3:2). The chemistry analog is high-temperature, short-duration reaction chemistry: kinetic products, many but loose.

4.3. G Dwarf Mechanism: The Dynamic Floor

G dwarfs fall in the intermediate regime: disk lifetimes of 2–5 Myr, moderate disk masses, moderate migration speeds. Neither slow enough for careful M-dwarf locking nor fast enough for F-dwarf kinetic capture. Resonant encounters are frequent but rarely successful: most are “near misses” where planets approach a resonance but cross through it before capture can complete.

The data are consistent with this picture. G dwarfs show the lowest bond fraction, the loosest bonds, and the narrowest bond palette. And crucially, the stability-filtered null shows that G-dwarf systems are indistinguishable from random packing within stability bounds.

4.4. The Deeper Pattern

The U-shape may be a general feature of systems where bond formation requires specific environmental conditions. Many chemical reaction systems show optimal-yield regimes—too cold and reactions don’t proceed; too hot and bonds break as fast as they form.

Planetary systems appear to show a similar pattern. M and F dwarfs both achieve bonding, but via different

Table 4. Change in bond fraction with stellar age. Young/old halves of the sample within each mass bin. Uncertainties are $\pm 1\sigma$ bootstrap confidence intervals.

Star type	N_{hosts}	Young half	Old half	Δ
M dwarf	33	$62.0 \pm 8.1\%$	$70.1 \pm 7.0\%$	+8.1%
K dwarf	105	$63.2 \pm 4.4\%$	$50.6 \pm 4.9\%$	−12.6%
G dwarf	105	$54.6 \pm 4.5\%$	$42.3 \pm 5.2\%$	−12.3%
F dwarf	48	$66.6 \pm 6.8\%$	$47.2 \pm 7.9\%$	−19.5%

routes (slow time vs. high energy density). G dwarfs are the intermediate case—neither cold-and-patient nor hot-and-aggressive. The Solar System sits at this point: bonded in places (Neptune-Pluto 3:2) but mostly unstructured.

5. AGE-BOND CORRELATION

If the two-mechanism picture is correct, bond fraction may evolve over time within each mass bin, though stellar age is not equivalent to disk lifetime and the mechanisms operating over Gyr timescales differ from those in the disk phase. Nonetheless, stellar age provides a complementary constraint.

We test this using stellar ages from the literature (Berger et al. 2018), available for 304 of 349 host stars in our sample. Stellar ages from galactic archaeology surveys carry significant uncertainties (typically ± 20 –30%), and our sample is heterogeneous, drawing from multiple surveys with different methodologies. The age results should be treated as suggestive rather than definitive.

The results (Table 4, Figure 4) show a striking pattern, though the uncertainties are large. Only K dwarfs show a statistically significant age-bond trend at 95% confidence ($\Delta = -12.6\%$, bootstrap CI $[-25.2\%, -0.2\%]$). The direction is consistent across all non-M types: bond fraction decays with age, most dramatically in F dwarfs (−19.5%), though the F-dwarf CI crosses zero (−39.7%, +1.0%).

M dwarfs are the only type where bond fraction increases with age (+8.1%), consistent with the picture that slow, continued dynamical evolution in M-dwarf systems preserves or strengthens resonant architectures over Gyr timescales. However, the M-dwarf CI is wide and crosses zero (−12.7%, +29.2%), so this should be treated as suggestive.

The age-bond correlation provides a tentative distinction between the two mechanisms. A “slow cooking” picture for M dwarfs predicts that bonds accumulate with time. A “fast chemistry” picture for F dwarfs predicts that bonds form rapidly in the disk phase and may decay afterward. The data are consistent with both pre-

Table 5. Adjacent planet pairs in the Solar System and their bond status. “Tightness” is the fractional deviation from the nearest integer ratio.

Pair	Period ratio	Nearest ratio	Tightness
Mercury–Venus	1.59	8:5	0.4% (near)
Venus–Earth	1.63	5:3	2.5% (near)
Earth–Mars	1.88	15:8	—
Mars–Jupiter	3.41	7:2	2.5% (near)*
Jupiter–Saturn	2.48	5:2	0.8% (loose)
Saturn–Uranus	2.85	20:7	—
Uranus–Neptune	1.88	15:8	—
Neptune–Pluto†	1.50	3:2	0.01% (clean)

dictions, but larger, more homogeneous age samples are needed to confirm.

6. THE SOLAR SYSTEM IN CONTEXT

Our Solar System orbits a G dwarf ($M_{\odot} = 1.0$). Table 5 summarizes its planetary bonds.

*Mars–Jupiter: period ratio computed as Jupiter’s period divided by Mars’, but the asteroid belt occupies the space between them, raising questions about “adjacency” in our framework.

†Including Pluto as a resonant Kuiper belt object, the Solar System contains one exceptionally clean 3:2 commensurability with Neptune; among the eight major planets, adjacent low-order commensurabilities are mostly loose or absent.

The Solar System has one near-perfect commensurability (Neptune–Pluto 3:2, the tightest in the entire dataset) and a few loose near-commensurabilities. This is consistent with the G-dwarf population mean: mostly non-bonded, occasional near-resonances, nothing close to the multi-resonance chain architectures of systems like TRAPPIST-1 or HD 110067.

The implication: the Solar System is not typical of the highly bonded architectures common in M- and F-hosted compact multi-planet systems. It is typical of the weakly bonded G-dwarf subset. Surveys that extrapolate planetary architecture statistics from the Solar System may systematically underestimate how structured planetary systems can be around other stellar types.

7. CONCLUSIONS

1. Of 846 adjacent planet pairs in 336 multi-planet systems, 56.6% are within $\pm 3\%$ of a low-order integer period ratio. Near-integer commensurability is the norm, not the exception.
2. The mean system-level bond fraction follows a U-shaped dependence on stellar mass: M dwarfs

Bond Fraction vs. Stellar Age — M dwarfs build bonds, others decay

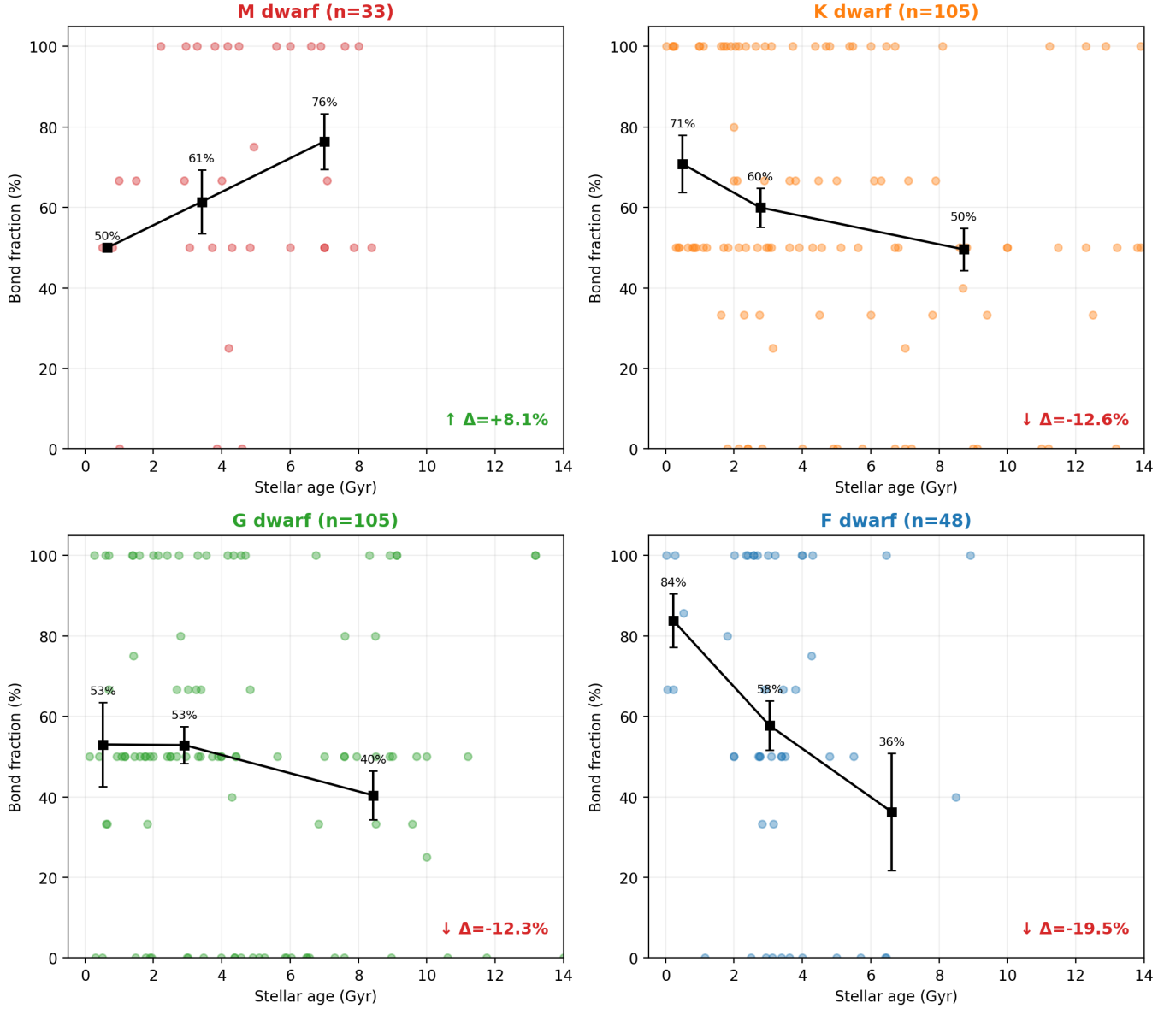


Figure 4. Bond fraction vs. stellar age, separated by stellar type. Black squares show age-binned averages with error bars. M dwarfs are unique: bond fraction increases with age ($\Delta \approx +8\%$), while other types show negative or flat trends.

(66.0%) and F dwarfs (56.0%) bond more strongly; G dwarfs (47.0%) bond the least.

3. A stability-filtered null ensemble shows that G dwarfs are at the dynamical floor: their bond fraction is indistinguishable from random packing within stable bounds (52.6% vs. 52.9%). M and F dwarfs show apparent excess bonding above the null.
4. M dwarfs produce the tightest commensurabilities (closest to integer ratios), consistent with long

post-capture settling times. G dwarfs produce the loosest bonds.

5. The U-shape is consistent with two candidate mechanisms: M-dwarf slow-capture (long disk lifetime, careful settling) and F-dwarf fast-capture (short disk lifetime, kinetic capture).
6. Age-bond correlations suggest the two mechanisms produce opposite evolutionary signatures: M dwarfs show tentatively increasing bond fraction with age (+8%); K, G, and F dwarfs show de-

creasing bond fraction with age (-12% to -20%). These results are preliminary and require homogeneous age catalogs to confirm.

7. The Solar System, orbiting a G dwarf at the bottom of the bonding curve, is typical of weakly bonded G-dwarf systems: mostly unstructured, with one clean commensurability and few tight bonds.

7.1. Predictions

- If the M-dwarf mechanism is correct, bond fraction should correlate with stellar age in M-dwarf systems (older $\rightarrow$ more settling $\rightarrow$ tighter bonds).
- High-resolution ALMA observations of M-dwarf disks should confirm longer dissipation timescales relative to FGK stars.
- Bond fraction around F dwarfs should anticorrelate with disk accretion rate (faster accretion $\rightarrow$ less settling).
- Direct imaging surveys of M-dwarf systems should preferentially find resonance chain architectures.

7.2. Future Work

- Population synthesis with disk evolution models to test the two-mechanism hypothesis quantitatively.
- N-body integrations comparing stability lifetimes of M-dwarf vs. G-dwarf resonance chains.
- Extension to non-adjacent pairs (1,3-interactions)—the gravitational analog of ring strain in molecular chemistry.
- Application of bond classification to the full Kepler DR25 catalog and upcoming TESS multi-planet systems.
- Homogeneous stellar age determination (e.g., via *Gaia* asteroseismology) to confirm the age-bond trends.

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REFERENCES

- Berger, T. A., Huber, D., Gaidos, E., & van Saders, J. L. 2018, *ApJ*, 866, 99, doi: [10.3847/1538-4357/aada83](https://doi.org/10.3847/1538-4357/aada83)
- Fabrycky, D. C., Lissauer, J. J., Ragozzine, D., et al. 2014, *ApJ*, 790, 146, doi: [10.1088/0004-637X/790/2/146](https://doi.org/10.1088/0004-637X/790/2/146)
- Fang, J., & Margot, J.-L. 2012, *ApJ*, 761, 92, doi: [10.1088/0004-637X/761/2/92](https://doi.org/10.1088/0004-637X/761/2/92)
- . 2013, *ApJ*, 767, 115, doi: [10.1088/0004-637X/767/2/115](https://doi.org/10.1088/0004-637X/767/2/115)
- Haisch, K. E., Lada, E. A., & Lada, C. J. 2001, *ApJL*, 553, L153, doi: [10.1086/320685](https://doi.org/10.1086/320685)
- Luger, R., Sestovic, M., Kruse, E., et al. 2017, *Nature Astronomy*, 1, 0129, doi: [10.1038/s41550-017-0129](https://doi.org/10.1038/s41550-017-0129)
- Millholland, S., Lissauer, J. J., Sullivan, G., et al. 2021, *ApJL*, 920, L14, doi: [10.3847/2041-8213/ac26b0](https://doi.org/10.3847/2041-8213/ac26b0)
- Mills, S. M., Fabrycky, D. C., Migaszewski, C., et al. 2016, *Nature*, 533, 509, doi: [10.1038/nature17445](https://doi.org/10.1038/nature17445)
- Pu, B., & Wu, Y. 2015, *ApJ*, 807, 44, doi: [10.1088/0004-637X/807/1/44](https://doi.org/10.1088/0004-637X/807/1/44)
- Sicilia-Aguilar, A., Henning, T., Hartmann, L., & MacGregor, M. 2010, *ApJ*, 710, 1067
- Weiss, L. M., Marcy, G. W., Petigura, E. A., et al. 2018, *AJ*, 155, 48, doi: [10.3847/1538-3881/aa9ff6](https://doi.org/10.3847/1538-3881/aa9ff6)
- Yasui, C., Kobayashi, N., Tokunaga, A. T., Saito, M., & Tokoku, C. 2010, *ApJ*, 723, 1686, doi: [10.1088/0004-637X/723/2/1686](https://doi.org/10.1088/0004-637X/723/2/1686)